EMMANUEL ABLAO

IT Specialist | Full-Stack Developer

Pililla, Rizal, Philippines | 09519044954 | emmanuelablao16@gmail.com

PROFILE

Full-Stack Developer and recent Information Technology graduate with hands-on, production experience building web, mobile, desktop, and IoT-integrated systems. Progressed from OJT intern to full-time IT Specialist within a single company after being absorbed for strong performance. Proficient in React, TypeScript, Node.js, and SQL/Supabase, with experience deploying live production applications (SSL/DNS configuration, Hostinger, GoDaddy) and integrating embedded hardware (ESP32, NFC) with backend APIs for real-time data processing. Adaptable and self-driven, with a continuous focus on learning modern web technologies and best practices.

SKILLS

Frontend: React, JavaScript, TypeScript, React Native
Backend & Database: Node.js, SQL, Supabase, Firebase
Desktop: Electron
IoT & Embedded: Internet of Things (IoT), PlatformIO
Tools & Version Control: Git, NPM
Security: JWT Authentication, AES-256-CBC Encryption, Row Level Security (RLS)

WORK EXPERIENCE

JJC Engineering Works & General Services — *Philippines*

IT Specialist (Absorbed from OJT — Full-Time) — Dec 2025 – Present

- Promoted from On-the-Job Trainee to full-time IT Specialist after graduation, having been absorbed into the company on the strength of consistent performance, technical initiative, and reliable delivery during the internship period.
- Continue to own and expand the company's internal SAP-style enterprise platform and Attendance Management System, taking on increasing responsibility for architecture decisions, feature scope, and production support.
- Serve as primary developer for internal tools spanning payroll/HR systems, attendance tracking, and IoT hardware integration used daily across HR, Finance, Procurement, and Operations departments.
- Provide ongoing maintenance, bug fixes, and feature enhancements for live systems relied upon by multiple departments and building entrances.

On-the-Job Trainee (OJT) — IT Department — Aug 18, 2025 – Dec 2025

- Joined as an OJT and was assigned to design and develop a full-stack SAP-style enterprise platform, replacing the company's manual Excel-based workflows for payroll, attendance logging, and total hours computation across all departments.
- Built a standalone Electron desktop application for barcode-based attendance capture, demonstrating the ability to independently deliver production-grade software during the internship.
- Delivered results that led to a full-time offer and absorption into the company immediately upon graduation, recognized for technical competence, ownership, and dependability.

PROFESSIONAL PROJECTS (LIVE, PRODUCTION SYSTEMS)

The following systems were designed and built for JJC Engineering Works & General Services and are actively deployed in production.

JJC Engineering — SAP System — *Philippines*

- Designed and developed a full-stack SAP-style enterprise platform for JJC Engineering, replacing manual Excel-based workflows for payroll, attendance logging, and total hours computation across all departments.
- Built a responsive web application with a public landing page and a secured employee dashboard featuring Announcements, Attendance time-in/out, a Cooperative Savings Program (Coop), and a Personal Information portal displaying employee benefits.
- Implemented a Department Selector routing system with dedicated modules for HR, Finance, Procurement, and Operations, enabling role-specific access and workflows per department.
- Secured all API communications using JWT-based authentication with AES-256-CBC symmetric encryption on all frontend requests and backend responses, protecting sensitive employee and payroll data end-to-end.
- Developed a companion cross-platform mobile application (React Native) mirroring all web features, with Firebase Cloud Messaging (FCM) push notifications to alert employees of new announcements and updates even when the app is closed.

JJC Attendance Management System — *Philippines*

- Built a standalone Electron desktop application (Node.js, better-sqlite3) that reads employee barcode IDs from a connected barcode scanner and transmits each scan to a central server for calculation and recordkeeping.
- Implemented a cache-first, offline-capable architecture that stores barcode scans in a local SQLite database when the network is down, then automatically syncs queued records to the server once connectivity is restored.
- Developed an in-app hours-computation engine that calculates regular and overtime hours per company time policy, covering multiple shift sessions (morning, afternoon, evening, overtime), grace periods, overnight-shift detection, and Sunday-specific schedule rules.
- Built a polling-based sync service with automatic retry and a daily fallback sync that detects employee changes in near real time, blocking clock in/out for newly added or admin-deactivated employees until verified at the admin office.

- Secured device-to-server communication with AES-256-CBC encrypted API requests, added duplicate-scan detection to prevent double entries, and shipped auto-update support for kiosks deployed across multiple building entrances.

INDEPENDENT PORTFOLIO PROJECTS

The following projects were self-initiated to independently practice and demonstrate technical skills across different platforms and domains.

ERJ Smart Solutions — *Philippines*

- Built an HR and attendance management platform supporting mobile app clock-in and dedicated desktop terminal clock-in, with real-time sync to a central system.
- Integrated custom hardware (ESP32 with PN532 NFC module) for physical attendance scanning, transmitting real-time data to the backend for processing.
- Designed a scalable, subscription-based architecture supporting multiple company accounts with tiered feature access.
- Built a centralized dashboard for department management, shift coordination, attendance tracking, and leave request handling.
- Implemented role-based access control (Admin, HR, Manager, Staff) with strict permission enforcement to protect sensitive workforce data.

Vue sur la Montagne — *Philippines*

- Developed a fully responsive single-page React application for a resort booking platform.
- Built custom animation hooks using Intersection Observer for scroll-based interactions and smooth page transitions.
- Designed a hero carousel with touch and keyboard navigation, integrated with a live booking form including date pickers and guest selectors.
- Built an interactive accommodations showcase with hover details, an amenities grid, and an animated testimonials gallery with lightbox.
- Implemented a validated contact form for guest inquiries.

Quick Track — *Philippines*

- Built a cross-platform mobile application for field technicians using React Native and Expo (iOS and Android).
- Implemented offline-first data syncing with local SQLite queueing, batched REST API uploads, and per-record retry logic.
- Developed equipment management features covering multiple categories (HVAC, Electrical, Plumbing, Safety, Mechanical) with serial number tracking and service logs.
- Built on-device reporting with camera integration and shareable PDF report generation.
- Implemented a push notification system for overdue equipment alerts and scheduled daily review reminders.

EAT-ALL POS — *Philippines*

- Developed a cross-platform desktop POS application using Electron and React, supporting Windows, macOS, and Linux.
- Built an offline-first architecture using SQLite for local data persistence, managed via IPC bridges between main and renderer processes.
- Designed a touch-optimized cashier interface with real-time VAT and discount calculations, multiple payment methods, and quick product search.
- Implemented real-time inventory tracking with automatic stock deduction on order completion.
- Built an analytics dashboard delivering daily and weekly sales insights, top-selling items, and low-stock alerts.

Inventory Control System — *Philippines*

- Built a full-stack inventory management application using React, Vite, and TailwindCSS with a Supabase (PostgreSQL) backend for centralized, secure data management.
- Implemented offline-first functionality using Service Workers and IndexedDB, enabling uninterrupted stock scanning and purchase order creation with automatic sync on reconnect.
- Built a real-time synced dashboard with Recharts, using Supabase real-time subscriptions to keep stock levels and alerts updated across all active sessions.
- Implemented multi-role access control (Admin, Manager, Staff) with Row Level Security (RLS), audit trails, automated purchase order generation, and CSV reporting.

Let's Play Song — *Philippines*

- Built a music player app for phones (iPhone and Android) that lets people stream and listen to songs anywhere, with a simple, easy-to-use design.
- Added the option to sign in with an email or a Google account, so each person's playlists and saved songs stay tied to their own account.
- Made it possible to create playlists that automatically stay up to date on every device, so a playlist saved on one phone shows up the same way on another.
- Added a built-in sound equalizer so listeners can adjust the audio to their own taste, like boosting the bass or making vocals clearer.
- Built a "download for offline" option so users can save their favorite songs directly on their phone and keep listening even without an internet connection.

EDUCATION

Bachelor of Science in Information Technology — *Antipolo City, Rizal | Sept 2022 – Dec 2025*
ICCT Colleges — Awarded Best in Website Development

BS Information Technology (Vocational, 2 yrs) — *Antipolo City, Rizal | May 2016 – Oct 2017*
Gateway Institute of Science and Information Technology